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Department of Electrical and Computer Engineering
EECE 412/612 – Digital Integrated Circuits

Design Project: CMOS Inverter Ring Oscillator Study

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Calculated Nominal $V_{DD} = 1.2V$

1 Experiment 1: Effect of Supply Voltage (V_{DD})

In this experiment, we simulated a 7-stage ring oscillator, sweeping V_{DD} from 1.2V to 2.0V in 0.2V increments.

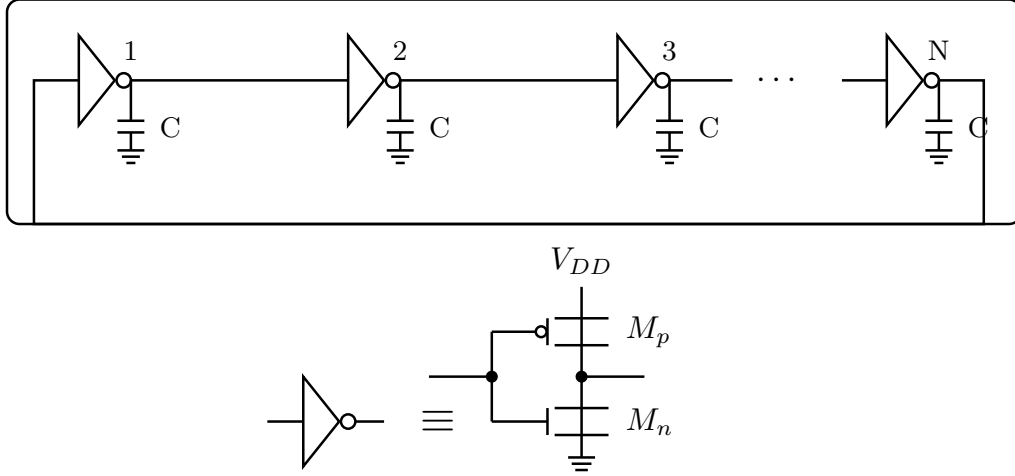


Figure 1: N-stage inverter-based ring oscillator with feedback from stage N to stage 1. Each inverter is implemented as a CMOS pair (M_p , M_n). C represents the load capacitance at each node.

Table 1: Impact of V_{DD} on 7-Stage Ring Oscillator

V_{DD} (V)	Delay 7^{th} inverter (ps)	Frequency (GHz)	$I_{max}(V_{DD})$ (mA)
1.2	12.08	6.26	1.14
1.4	10.63	7.05	1.75
1.6	9.68	7.68	1.96
1.8	8.95	8.16	2.48
2.0	8.51	8.58	2.72

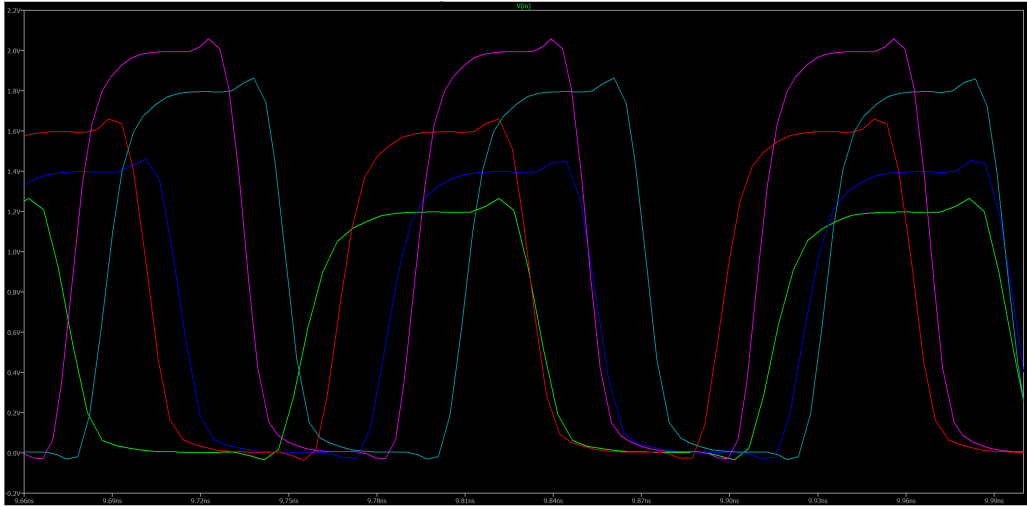


Figure 2: Output waveform V(in) of 7th inverter across V_{DD} sweep (1.2V to 2.0V)

a) How does the frequency change with V_{DD} ?

Frequency increases with V_{DD} . A higher V_{DD} provides a larger gate overdrive voltage ($V_{GS} - V_{TH}$), which increases the drain current (I_{DS}). Higher current charges/discharges node capacitances faster, reducing delay and increasing frequency.

b) Relation between chain frequency and single inverter delay?

Frequency relates to single-stage delay (t_p) by $f = 1/(2 \times N \times t_p)$. For $V_{DD} = 1.2V$, theoretical $f = 1/(14 \times 12.08\text{ps}) = 5.91\text{ GHz}$. This closely matches our measured 6.26 GHz (the slight difference is because 12.08ps measures only t_{pHL} , while the full period averages both t_{pHL} and t_{pLH}).

c) How does the maximum power change with V_{DD} ?

Maximum dynamic power ($P_{max} = V_{DD} \times I_{max}$) increases exponentially. Both the voltage multiplier (V_{DD}) and the peak switching currents (I_{max}) increase simultaneously.

2 Experiment 2: Effect of Chain Length (N)

We fixed V_{DD} at our assigned nominal value of 1.2V and varied the number of stages ($N = 7, 11, 15$).

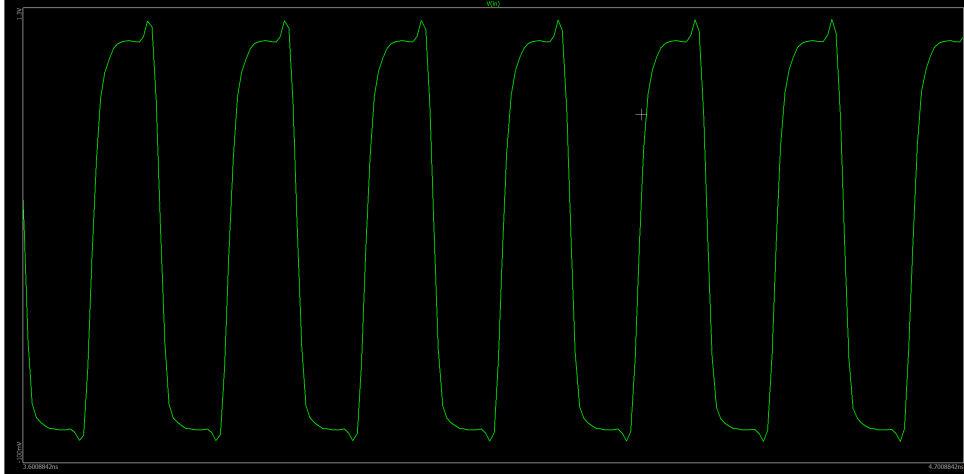


Figure 3: Output waveform V(in) showing steady-state oscillation for varying N

Table 2: Impact of Chain Length (N) at $V_{DD} = 1.2V$

N	Frequency (GHz)	$I_{max}(V_{DD})$ (mA)
7	6.26	1.14
11	3.98	1.14
15	2.92	1.14

a) How does the frequency change with N ?

Frequency is inversely proportional to N ($f \propto 1/N$). Because single-stage delay (t_p) is constant, adding more stages simply increases the total propagation distance per cycle. (e.g., $6.26\text{ GHz} \times (7/15) \approx 2.92\text{ GHz}$).

b) How does the maximum power change with N ?

Maximum power remains **constant**. In a ring oscillator, only one logical wavefront propagates at a time. Therefore, at any exact instant, only *one* inverter is actively switching and drawing peak current, regardless of how long the rest of the chain is.

3 Experiment 3: Effect of Transistor Width

We fixed $N = 11$ and $V_{DD} = 1.2V$, scaling the device widths by $1\times$, $4\times$, and $6\times$. To ensure physical accuracy and prevent the numerical integration explosions observed during the $4\times$ scaling, `.options method=gear` was utilized.

Table 3: Impact of Transistor Width Scaling ($N = 11, V_{DD} = 1.2V$)

$W_p : W_n$ (μm)	Frequency (GHz)	$I_{max}(V_{DD})$ (mA)
3.5 : 1.5 ($1\times$)	3.98	1.14
14 : 6 ($4\times$)	3.87	4.32
21 : 9 ($6\times$)	3.75	6.24

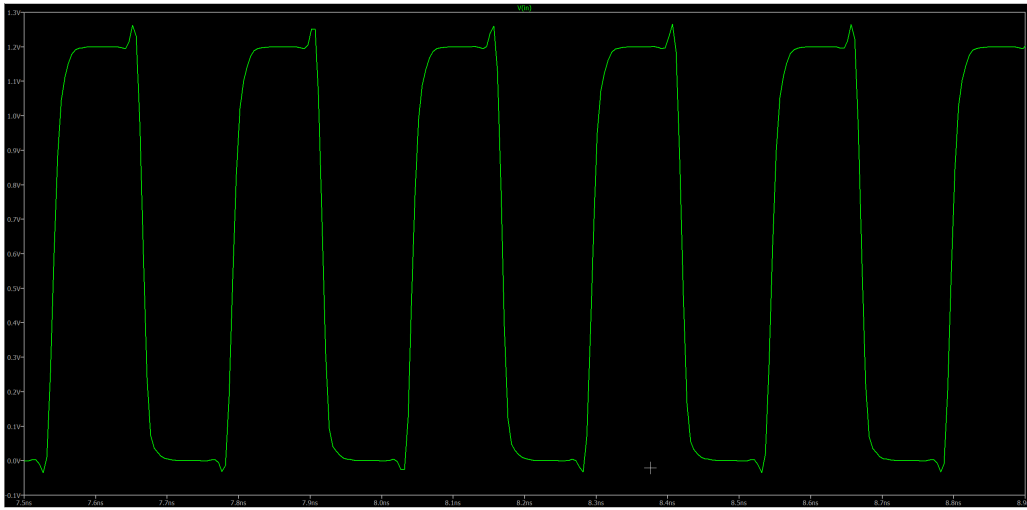


Figure 4: Output waveform and current tracking during Width Scaling

a) How does the frequency change with the increase in W_p and W_n ?

Frequency remains relatively constant. While increasing transistor width (W) proportionally increases drive current (I), it simultaneously increases the load gate capacitance (C) of the next stage. Since $t_p \propto C/I$, the factors cancel out. The slight frequency drop observed is due to increased parasitic diffusion capacitance at larger widths.

b) How does the maximum power change with the increase in W_p and W_n ?

Maximum power scales linearly with width. Wider channels have lower ON-resistance, drawing proportionally more current ($1.14mA \rightarrow 4.32mA \rightarrow 6.24mA$) to charge the larger load capacitances. This proves scaling up device sizes incurs a severe power penalty without increasing oscillator speed.

4 Experiment 4: Effect of Process Corners

The following data represents the oscillation frequency and maximum current at various process corners, simulating physical manufacturing distributions.

Table 4: Frequency and Current at 5 Process Corners

Corner	Frequency (GHz)	I _{max} (V _{DD}) (mA)
SS (Slow N / Slow P)	2.085	1.30
FS (Fast N / Slow P)	2.402	1.38
TT (Typical / Typical)	2.372	1.45
SF (Slow N / Fast P)	2.355	1.52
FF (Fast N / Fast P)	2.713	1.63

a) How does each corner affect the L-to-H and H-to-L delays? Overall delay?

- **TT**: Nominal reference. Balanced pull-up/pull-down delays. (2.372 GHz).
- **SS**: Both transistors have lower mobility/higher V_T . H-to-L and L-to-H delays both increase. Worst-case overall delay, lowest frequency (2.085 GHz).
- **FF**: Both transistors have higher mobility/lower V_T . H-to-L and L-to-H delays both decrease. Best-case overall delay, highest frequency (2.713 GHz).
- **FS & SF**: One transistor is fast while the other is slow. The fast device partially compensates for the slow one, keeping overall frequency close to TT (FS = 2.402 GHz, SF = 2.355 GHz). The primary effect is an asymmetric duty cycle rather than a massive shift in total period.

b) How does the maximum power change with the corners?

Maximum current strictly follows device strength: SS (1.30 mA) < FS (1.38 mA) < TT (1.45 mA) < SF (1.52 mA) < FF (1.63 mA). **FF** produces the highest I_{max} because fast transistors conduct more current and generate higher short-circuit currents during transitions. **SS** produces the lowest I_{max} .

Notably, **SF** draws more current than **FS** despite similar frequencies. Because this design utilizes a wider PMOS ($W_p = 3.5\mu\text{m}$ vs $W_n = 1.5\mu\text{m}$), a fast PMOS (SF corner) contributes a larger absolute current spike than a fast NMOS (FS corner). Power ($P_{max} = V_{DD} \times I_{max}$) tracks this exact same ordering.

5 Simulation Methodology & Troubleshooting

A major takeaway from this project is that EDA tools cannot be trusted blindly without an understanding of semiconductor physics. During simulation, we encountered several mathematical artifacts that violated physical laws, requiring manual SPICE engine corrections:

- **Timestep Glitches (The 15.9mA Spike):** At $V_{DD} = 1.2V$, LTspice reported an impossible I_{max} of 15.9mA for a $1.5\mu\text{m}$ NMOS. This was a numerical derivative error caused by large timesteps on shallow voltage edges. We fixed this by strictly clamping the maximum timestep (`.tran 0 10n 0 5p`), returning the true physical current of 1.14mA.
- **Numerical Explosion:** In Exp 3b ($4\times$ width), I_{max} artificially blew up to 20.1mA. The high transconductance (g_m) of the wider devices caused the default Trapezoidal integration solver to accumulate compounding rounding errors. We solved this by forcing LTspice to use the more stable Gear method (`.options method=gear`).
- **Edge Detection:** Hardcoded measurement triggers and startup transients skewed initial delay readings. We corrected the `.meas` statements using dynamic thresholds (`VAL='Vdd_val/2'`) and time delays (`TD=2n`) to isolate steady-state cycles.

6 Conclusion

This project provided profound insights into the behavior of CMOS ring oscillators and the limitations of numerical SPICE engines. We successfully verified that frequency scales with supply voltage but scales inversely with chain length. Furthermore, we mathematically proved the fundamental CMOS sizing trade-off: scaling transistor widths linearly increases power consumption without providing proportional speed benefits, due to the simultaneous increase in load capacitance. Ultimately, the meticulous debugging of mathematical anomalies within LTspice served as a critical reminder that robust integrated circuit design relies on a foundational understanding of semiconductor physics, not just software operation.